



Coupling of vegetation filter strips and biochar reduced the loss of nitrogen and phosphorus nutrients in subtropical riparian areas under continuous rainfall conditions

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ARTICLE INFO

Keywords:

Biochar
Vegetative filter strips
Non-point source pollution
Runoff

ABSTRACT

The vegetation filter strip (VFS) in riparian zones serves as an ecological barrier to prevent nutrient inflow from farmland to rivers, but its efficacy diminishes under continuous heavy rainfall. Using flume experiments simulating three continuous rainfall events, this study examined the effects of VFS alone and combined with biochar application rates (30, 60, 90 t ha⁻¹) on nutrient loss dynamics, specifically total nitrogen (TN) and total phosphorus (TP). Results indicated that VFS alone reduced nutrient loads in surface runoff, though its efficiency declined with repeated rainfall. TN and TP reduction decreased from 38.0 % to 14.0 % and from 33.2 % to 13.9 %, respectively. Biochar application alone at low to medium rates reduced runoff and nutrient concentrations, while high biochar rates increased runoff but significantly reduced nutrient concentrations. The combined use of VFS and biochar achieved the highest nutrient reduction and maintained stability across rainfall events. Applying 30 t ha⁻¹ biochar with VFS reduced TN by 59.9 %, 56.0 %, and 45.0 % over three events, while 90 t ha⁻¹ biochar reduced TP by 62.5 %, 64.8 %, and 75.7 %. Subsurface runoff nutrient concentrations fluctuated but stabilized with the combined measures, especially during prolonged rainfall. Notably, while VFS-biochar coupling reduced SSR, its effect was less pronounced than biochar alone in surface runoff control due to root-induced macropore flow. This technology provides an efficient and stable solution for the treatment of agricultural non-point source pollution in subtropical riparian zone, and has broad application prospects in the field of degraded soil remediation.

1. Introduction

Water resources are vital for the survival and development of organisms, necessitating rigorous environmental pollution controls to preserve water body purity (Zhou et al., 2023). Post-Green Revolution agricultural practices, particularly the overuse of chemical fertilizers and pesticides, have led to their infiltration into aquatic systems via runoff and leaching, severely polluting surface and

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groundwater sources such as lakes and streams, and exacerbating the issue of agricultural non-point source (NPS) pollution. For instance, the US Environmental Protection Agency has identified NPS pollution as a primary contributor to river and stream degradation, with agricultural sources being among the top six culprits (Brown and Froemke, 2012). Similarly, in the European Union and Germany, agriculture significantly contributes to the contamination of surface waters with new psychoactive substances, accounting for approximately 55 % and 48 % of such pollutants, respectively (Volk et al., 2009). In China, agricultural discharge is responsible for 46.5 % of total nitrogen (TN) and 67.2 % of total phosphorus (TP) loads in overall wastewater emissions (Zhou et al., 2023). Addressing nutrient pollution in agriculture remains a formidable challenge due to the extensive number of stakeholders involved and the spatial and temporal variability in nutrient loading.

As an important ecological link between terrestrial and aquatic ecosystems, the vegetation filter strips (VFS) in the riparian zone of the river significantly reduces emissions of pollutants such as TN and TP through physical, chemical and biological processes, such as interception, filtration, deposition, degradation and absorption (Mayer et al., 2007; Lind et al., 2019; Oshunsanya et al., 2019). However, the effectiveness of VFS in reducing NPS pollution is affected by seasonal conditions and rainfall patterns (Hou et al., 2022). For example, in the subtropical red soil farming area of China, the rainy season overlaps with the farming season, resulting in frequent rainfall and high nutrient load in runoff. Despite the existence of VFS, strong and short-duration rainfall events can cause significant nutrient losses. Additionally, long-term deployment of VFS can result in a substantial accumulation of nitrogen and phosphorus in surface soil, thereby diminishing the effectiveness of VFS in reducing these nutrient losses (Xia et al., 2015). The effectiveness of VFS is also affected by its width, and a study by (Dindaroğlu et al. (2015) recommended a minimum width of 3–10 m for VFS. In small watershed areas where VFS cannot be implemented due to constraints such as basic agricultural land or where the setup width falls short of the recommended value (Sun et al., 2013), alternative measures or a combination with VFS may be necessary to effectively minimize nutrient losses.

Biochar, derived from the pyrolysis of agricultural residues under high temperatures and limited oxygen conditions (Nguyen et al., 2019), presents innovative applications in soil amendment and pollution control. Its unique properties, such as a porous structure, extensive surface area, and abundant functional groups, enable strong ion adsorption capabilities (Peng et al., 2019; Wu et al., 2021). These characteristics make biochar an effective tool in mitigating nitrogen and phosphorus runoff, thus offering substantial potential for enhancing soil quality and reducing agricultural pollution (Yin et al., 2023). Meanwhile, the stable adsorption properties and chemical reactivity of biochar allow it to bind with organic matter and minerals, thus reducing nutrient losses through leaching and surface runoff (Palansooriya et al., 2019). Biochar is enriched in soil aggregates, and the protection of soil aggregates is beneficial to the long-term retention of nitrogen and phosphorus (Lee et al., 2018), which provides a basis for biochar to reduce agricultural non-point source pollution. In addition, the researchers also paid attention to the effect of the combination of biochar and vegetation on vegetation growth. Biochar has a positive effect on vegetation development. It can enhance the absorption of soil moisture and nutrients by plants, thus providing the best environment for vegetation growth and promoting plant growth (Kuttner and Thomas, 2017; Adak et al., 2020; Czarnes et al., 2000). Combined with the adsorption of biochar, these findings can prove the applicability of the coupling effect of biochar and vegetation in the field of pollution reduction. These effects may include the interaction between biochar and root growth, and the adsorption of nutrients in runoff by the interaction between biochar and vegetation (Zhao et al., 2023). In the joint study of biochar and vegetation, most researchers pay attention to the effect of biochar on vegetation growth, and the existing research on the blocking effect of vegetation and biochar on runoff nutrients is not clear. Therefore, it is necessary to understand the effects of vegetation and biochar on NPS.

The process of nutrient loss caused by agricultural NPS pollution is very dynamic and complex, which is affected by many factors such as vegetation coverage, agricultural practice, rainfall characteristics and topography (García-Díaz et al., 2017). Among these factors, rainfall characteristics are particularly important (Wu et al., 2018). Relevant studies have shown that rainfall events with high intensity, short duration and high frequency tend to cause the largest runoff loss and the largest chemical pollution (Römken et al., 2002; Liu et al., 2014). Rainfall and runoff, as two important driving forces affecting runoff, both affect the migration ability of nutrients in runoff. During rainfall events in the riparian zone, it is crucial to consider the dynamics of soil nutrient separation and transportation due to raindrop splashing (Xu et al., 2019). In addition, the inflow of water and nutrients from the upper farmland into the buffer zone increases the load on the riparian area, which challenges its role as an effective nutrient barrier between farmland and adjacent water bodies. At the same time, many studies have focused on the surface runoff in the riparian zone, but the exploration of interflow is still insufficient, especially under the influence of uphill water inflow. Our study fills this gap by exploring how rainfall and catchments interact with biochar and vegetation filter strips, making it possible to redefine best practices for pollution control and water quality management in agricultural landscapes.

Rainfall simulation is a useful method for measuring water, sediment and nutrient loss because it allows for repeated experiments and accurate control of rainfall intensity and distribution (Iserloh et al., 2013), to quantify the various phenomena and problems associated with rainfall. For example, Wang et al. (2018) studied the effects of hedgerows on atrazine runoff and nutrients under simulated rainfall. (Li and Pan, 2020) simulated the dynamic properties of runoff erosion on steep slopes with forage under rainfall and inflow conditions. Different soil management strategies for controlling non-point source pollution in the riverside area are deployed in the laboratory, and the specific simulation of continuous rainfall characteristics and upper farmland nutrient characteristics can be used as a useful tool to compare the loss of nitrogen and phosphorus under different soil management strategies. Through rainfall simulation experiments, this study aims to: (1) quantify the combined effects of continuous rainfall and upslope inflow on nutrient concentration and load characteristics in riparian zones; (2) investigate the effect of VFS or different biochar application rates on runoff nutrient reduction under single and coupled effects; (3) explore the optimal application ratio of biochar within VFS, providing a reference in reducing non-point source pollution. The study provides an efficient and stable solution for nutrient loss control under continuous rainfall through VFS-biochar synergistic technology.

2. Materials and methods

2.1. Experimental site and materials

The experimental soil was collected from Huanjiang County, a subtropical monsoon climate zone in Southwest China. The annual average temperature in this area is 19.9 °C. May to September is the main month of annual precipitation, with an annual precipitation of 1455.6 mm. For this study, red soil was extracted from the top 40 cm of the soil profile. According to the USDA texture classification, this soil is categorized as Sandy Loam (7.80 % clay, 50.59 % silt, 41.61 % sand), with a soil bulk density of 1.36 g cm⁻³. The concentrations of TN and TP in soil were 74.67 mg / kg and 33.34 mg / kg, respectively. To prepare the samples, the soil was air-dried under natural conditions until the moisture content dropped below 10 % (Table 1). Subsequently, the dried soil was sieved through a 10-mm mesh to eliminate larger particles, such as stones, organic debris, and plant roots.

The experiment utilized corn stalks as the raw material, pyrolyzing them at a high temperature of 400 °C to produce biochar. The concentrations of TN and TP in biochar were 219.21 mg / kg and 91.58 mg / kg, respectively. The use of commercial biochar was primarily to ensure that the results have broad practicality and representativeness (Zhang et al., 2019). Therefore, the biochar used in the experiments was bought from Lize Environmental Protection Technology Co., Ltd., Henan Province, China. Additionally, commercial biochar is readily available. The biochar was crushed and sieved through a 2 mm mesh, then uniformly mixed with the test soil before the experiment. That is because biochar particles smaller than 2 mm are widely used in practice, in order to avoid systematic errors caused by biochar particle size (Yin et al., 2023).

Based on a comprehensive field survey of the basin, Bermuda grass (*Cynodon dactylon* L.), a prevalent herbaceous species in the subtropical region, was chosen to represent vegetation typically found on VFS. This species is common across subtropical zones, flourishing in areas near villages, along roadsides and riverbanks, and on uncultivated slopes (Noor et al., 2023). Prior to planting, the soil was thoroughly rinsed with clean water to remove any residues. The grass was planted with a target coverage of over 90 %, mimicking the natural growth conditions observed during field surveys. During the establishment phase, daily watering was conducted to support the initial growth, with equivalent irrigation provided to control groups to maintain consistent experimental conditions. Once bermudagrass thrives and the rhizomes penetrate the soil, they are considered mature and ready for further experiments.

2.2. Experimental design

To determine the universality of NPS, we set the original bare slope without any measures as the control group (CK). Three biochar application levels were set: 30 t ha⁻¹, 60 t ha⁻¹, and 90 t ha⁻¹. Based on different biochar application rates, a total of seven treatments groups were set up (Table 2), including single application of biochar (T30, T60, T90), only VFS (VFS) and VFS + biochar (VFST30, VFST60, VFST90). Each treatment was repeated three times. We scheduled three continuous rainfall events (Event 1, Event 2, Event 3), with a 24-hour interval between two adjacent rainfall events. This approach was to ensure that the soil's initial moisture content was as similar as possible across the experiments, thereby reducing systematic errors. The experimental design consists of two main components: an experimental flume and a runoff distribution tank (Fig. 1). Each flume is constructed with dimensions of 200 × 100 × 40 cm (length × width × depth). We set the slope to 15° in this study. At the lower end of each flume, two V-shaped collection channels are installed to direct runoff. Runoff buckets are strategically placed beneath the outlets of these channels to capture both surface runoff (SR) and subsurface runoff (SSR). To mitigate boundary effects, Vaseline is applied along the edges of the steel plates that form the flume sides.

For soil preparation, the soil is first sieved using a 10-mm mesh and then filled into the flume in successive layers. Compaction is conducted in four stages, ensuring a uniform total soil thickness of 40 cm. Biochar is incorporated into the soil at a depth of 0–30 cm. During the soil filling process, a custom-made iron plate is utilized to uniformly compact the soil layers, paying particular attention to the boundaries to ensure consistent compaction across the entire flume.

To replicate the conditions of upslope inflow, a specially designed surface runoff distribution system was implemented. The system includes a 500 L polyethylene (PE) water storage tank, which supplies water to the experimental setup. Flow rate control is achieved using an adjustable valve, which is linked to a flow meter for precise monitoring. At the junction between the flow meter and the soil flume, a steady-flow groove, measuring 40 cm × 100 cm × 30 cm and constructed from steel plates, is installed. Inside this groove, staggered baffles are placed to stabilize the water head, ensuring consistent flow. Water from the steady-flow groove is then directed onto a runoff dispersion plate, which evenly distributes the runoff as it enters the flume, guaranteeing a uniform and steady flow throughout the experimental process (Ao et al., 2024).

Table 1

The properties of the soil and the changes in soil properties after the application of biochar at different rates.

Properties	CK	30t/ha	60t/ha	90t/ha
Sand (%)	7.80 ± 0.15	7.99 ± 0.73	8.14 ± 0.87	8.30 ± 0.86
Silt (%)	50.59 ± 1.22	57.660 ± 2.98	59.346 ± 2.45	61.117 ± 3.39
Clay (%)	41.61 ± 1.28	34.353 ± 3.71	32.513 ± 3.32	30.587 ± 4.23
pH (1:1)	6.80 ± 0.07	6.98 ± 0.11	7.17 ± 0.06	7.39 ± 0.15
OM(%)	2.43 ± 0.35	2.48 ± 0.16	2.65 ± 0.22	2.79 ± 0.43
OC(%)	1.99 ± 0.33	2.79 ± 0.25	3.49 ± 0.25	3.79 ± 0.47
CEC (meg/100 g)	23.69 ± 0.14	23.12 ± 0.24	22.78 ± 0.24	22.40 ± 0.33

Table 2
Experimental treatments design table.

Treatment	Biochar application rate (t ha ⁻¹)	Vegetated surface	Continuous rainfall events	Concentration of contaminant (mg L ⁻¹)	
				TN	TP
CK	0	N	Event 1	3.0	0.5
			Event 2	1.8	0.4
			Event 3	1.2	0.3
T30	30	N	Event 1	3.0	0.5
			Event 2	1.8	0.4
			Event 3	1.2	0.3
T60	60	N	Event 1	3.0	0.5
			Event 2	1.8	0.4
			Event 3	1.2	0.3
T90	90	N	Event 1	3.0	0.5
			Event 2	1.8	0.4
			Event 3	1.2	0.3
VFS	0	Y	Event 1	3.0	0.5
			Event 2	1.8	0.4
			Event 3	1.2	0.3
VFST30	30	Y	Event 1	3.0	0.5
			Event 2	1.8	0.4
			Event 3	1.2	0.3
VFST60	60	Y	Event 1	3.0	0.5
			Event 2	1.8	0.4
			Event 3	1.2	0.3
VFST90	90	Y	Event 1	3.0	0.5
			Event 2	1.8	0.4
			Event 3	1.2	0.3

NOTE: CK, control group; T30, application of 30 t ha⁻¹ biochar alone; T60, application of 60 t ha⁻¹ biochar alone; T90, application of 90 t ha⁻¹ biochar alone; VFS, vegetative filter strip; VFST30, application of 30 t ha⁻¹ biochar in vegetative filter strip; VFST60, application of 60 t ha⁻¹ biochar in vegetative filter strip; VFST90, application of 90 t ha⁻¹ biochar in vegetative filter strip. N: no vegetation cover; Y: covered with vegetation.

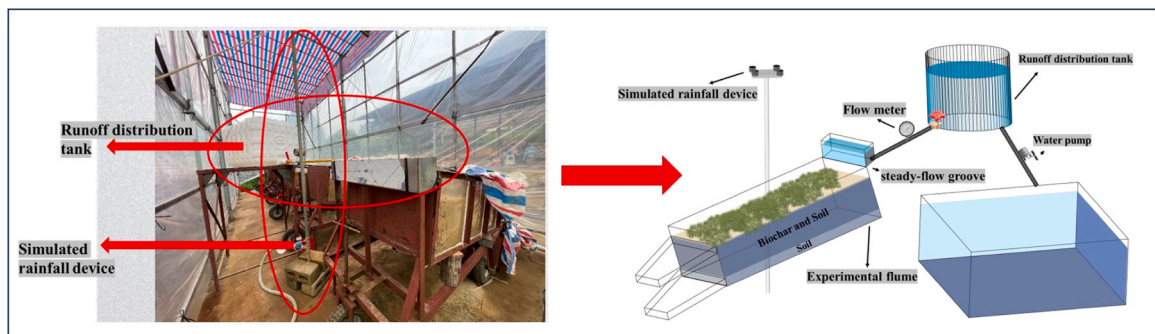


Fig. 1. Simulated rainfall field and experimental device schematic diagram.

The rainfall intensity for the experiment was set to 60 mm h⁻¹, with a duration of 60 min. A top-spraying rainfall simulation device, equipped with four US SPRACO conical nozzles, was used to generate the artificial rainfall. The device was positioned at a height of 4.75 m, and the system was calibrated before the experiment to ensure a uniform distribution of rainfall, achieving over 80 % uniformity within an effective radius of 2 m.

To prepare the soil for the experiment, a pre-rainfall event was conducted one day prior to the official test. This pre-rainfall was applied at an intensity of 30 mm h⁻¹ until surface runoff was observed, which brought the surface soil moisture to levels like those expected during actual rainfall conditions. Following the pre-rainfall, the experimental flume, now pre-moistened, was covered with plastic sheets and allowed to equilibrate for 24 h. This step ensured even moisture penetration and consistent soil conditions at the start of the main experiment.

The upslope flow is derived from the soil conservation service curve number (SCS-CN), which is a rainfall-runoff empirical model with assumptions and a small amount of input to estimate runoff caused by rainfall events (Cai et al., 2020). Considering parameters such as the area of the farmland watershed above the study zone and the amount of rainfall, we have set the inflow water volume for each rainfall event at 4 L min⁻¹, which is the catchment area within 10 m above the simulation.

During actual rainfall and upslope inflow processes, SR is collected every 2 min, and SSR is collected every 5 min, with each collection immediately weighed using an electronic scale. After allowing the runoff sediment samples to settle for 24 h, the supernatant is poured off, and the samples are dried in an oven at 105°C for 24 h until a constant weight is reached. The sediment weight is then

measured using an electronic balance.

Based on the current situation of non-point source pollution in red soil area of southern China and (Zhang et al., 2020) on the study of fertilizer loss in red soil farmland on various days of continuous rainfall, according to the different rainfall events, a certain concentration gradient of TP and TN mixture was configured to simulate surface runoff, and the concentration was determined by potassium persulfate anti-oxidation molybdenum antimony method. The total nutrient load of runoff released by each experimental flume is the sum of runoff releases at all times:

$$Ei = \sum_{j=1}^n (A_{PR(i,j)} \bullet C_{PR(i,j)}) \bullet \rho^{-1}$$

where Ei (g) is the pollutant load released from runoff i (i is SR). $A_{PR(i,j)}$ is the concentration of TN and TP of runoff i collected in a single time (mg L^{-1}); $C_{PR(i,j)}$ is the weight of runoff i collected in a single time (g); ρ is the density of rainwater, which is approximately 1000 mg L^{-1} ; and n represents the number of data collection events.

The load reduction efficiency (RE) of each test tank is the sum of the total runoff:

$$\text{reduction efficiency}(RE) = (1 - Ei \bullet E_{CK}^{-1}) \bullet 100\%$$

where E_{CK} is the total load of pollutants released by runoff in the control group.

In the process of actual rainfall, surface runoff (SR) was collected once every 2 min, and subsurface runoff (SSR) was collected once every 5 min. After the experiment, the inflow water samples were collected to determine the water quality indicators. The collected samples were weighed to calculate the flow rate. The collected water samples are transported back to the laboratory and immediately determined or stored at $< 4^\circ\text{C}$.

2.3. Scanning electron microscopy

Aggregates ranging in size from 0.5 to 2 millimeters were carefully selected from the prepared soil, biochar, and soil samples treated with varying biochar application rates using plastic tweezers. These selected aggregates were then affixed directly to conductive adhesive and underwent gold coating for 45 s at a current of 10 mA using a Quorum SC7620 sputter coater. After coating, the samples were analyzed using a TESCAN MIRA LMS Scanning Electron Microscope (SEM) equipped with Energy Dispersive X-ray Spectroscopy (EDS). EDS was employed to determine the elemental composition within the microstructure of the aggregates, while

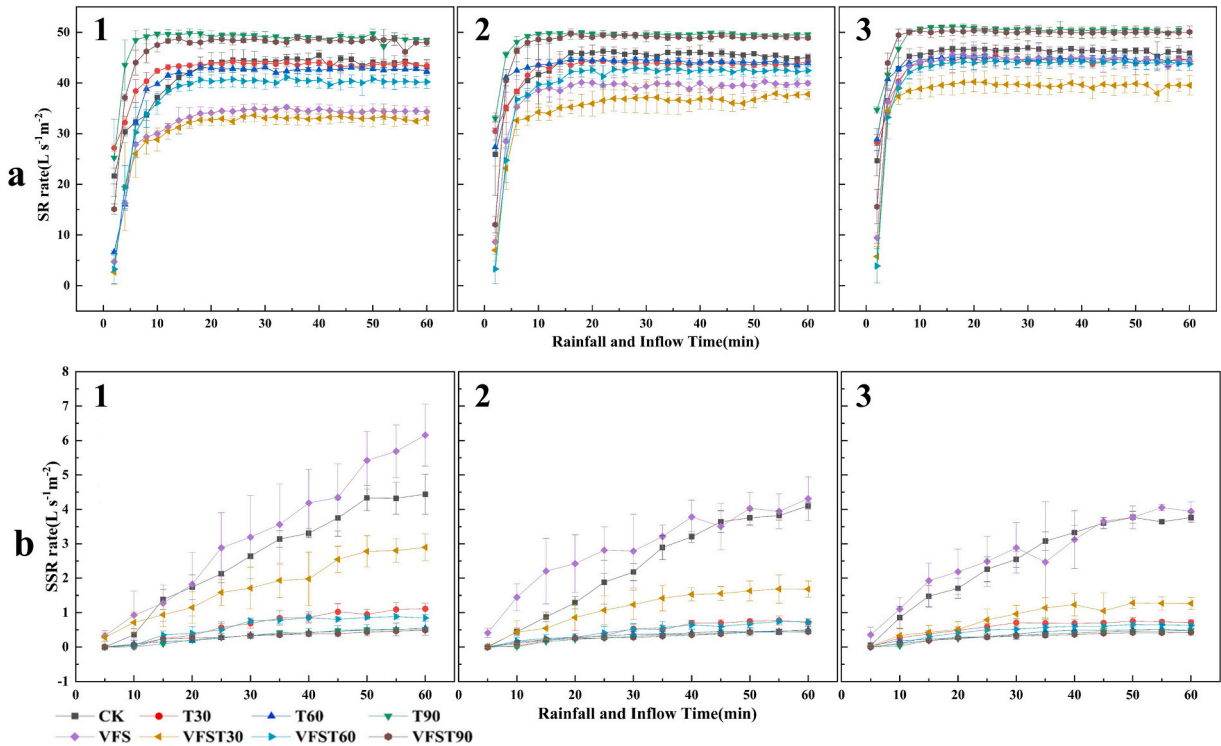


Fig. 2. The rate of surface runoff (SR) (a) and subsurface runoff (SSR) (b) under three consecutive rainfall and inflow events for Vegetative Filter Strips (VFS), individual application of biochar(T), and the coupled impact of Vegetative Filter Strips and biochar (VFST). The average of three replicates is shown with standard deviation bars. 1, 2 and 3 represent the first, second and third rainfall and inflow events respectively.

backscattered electron imaging provided black and white images that depicted the average atomic abundance within the samples.

2.4. Data analyses

One-way analysis of variance (ANOVA) and least significant difference (LSD) post hoc test were used to analyze the significant levels of SR, SF and sediment yield among different treatments ($P = 0.05$). Regression analysis was used to determine the relationship between biochar application rate, vegetation filter belt and soil erosion. Partial least squares structural equation modeling (PLS-SEM) is used to determine the coupling effect of rainfall and upstream inflow events, which provides two fitting indexes: standardized root mean square residual (SRMR) and normalized fitting index (NFI). These indicators are essential for evaluating the goodness of fit of the PLS-SEM model. When the SRMR value is less than 0.08 and the NFI is greater than 0.90, the model is considered to have a good fit. All statistical analyses were performed using SPSS 26. Origin 2021 was used for graphics drawing and data fitting, and the data of runoff and sediment yield were processed by Excel 2016.

3. Results

3.1. Runoff characteristic

Compared with (CK), VFS significantly reduced SR, but with the increase of rainfall events, the SR rate of VFS was close to that of CK (Fig. 2a). By the third event, the maximum difference in runoff rate between VFS and CK at the same time was only $2.0 \text{ L s}^{-1} \text{ m}^{-2}$. For treatments with biochar alone, T90 consistently exhibits higher SR rates than CK across all events, with an average increase of 12.2 %. In contrast, T30 and T60 show a decrease in SR rates relative to CK as the number of rainfall events increases. Meanwhile, VFST90 maintains a consistently higher SR rate compared to CK, with an average increase of 28.0 %. VFST30 remains consistently lower across all events, with an average decrease of 7.7 %. The SR rate of VFST60 increased the fastest in the early stage of rainfall, and its maximum SR growth rate was 182.0 %.

VFS reduced subsurface runoff (SSR), and with the increase of the number of events, it gradually approached the SSR rate of CK (Fig. 2b). Biochar application alone effectively reduces and SSR rates, and T60 and T90 showed similar trends in flow production across all events. The addition of biochar to VFS also reduced the SSR rate. It is worth noting that the SSR of VFST90 decreased the most, and its maximum SF rate phase decreased by $5.7 \text{ L s}^{-1} \text{ m}^{-2}$, $3.9 \text{ L s}^{-1} \text{ m}^{-2}$ and $3.6 \text{ L s}^{-1} \text{ m}^{-2}$, respectively.

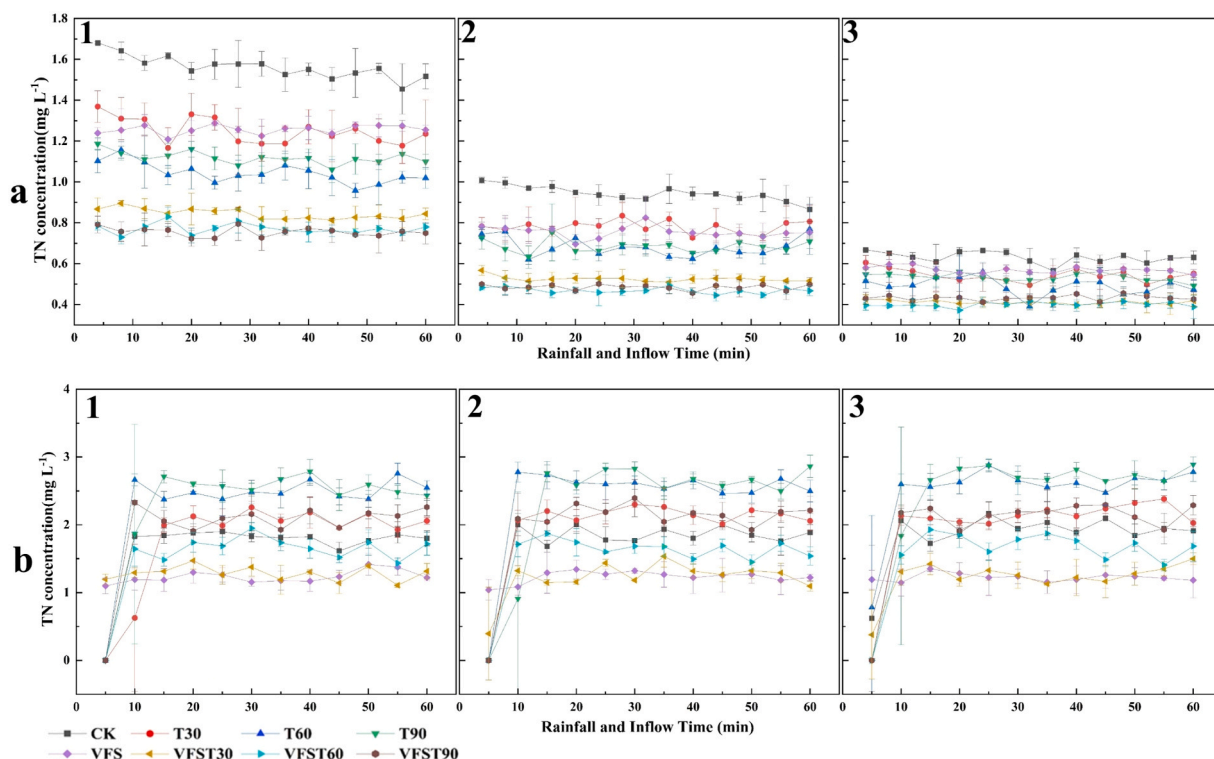


Fig. 3. The TN concentration of surface runoff (SR) (a) and subsurface runoff (SSR) (b) under three consecutive rainfall and inflow events for Vegetative Filter Strips (VFS), individual application of biochar(T), and the coupled impact of Vegetative Filter Strips and biochar (VFST). The average of three replicates is shown with standard deviation bar. 1, 2 and 3 represent the first, second and third rainfall and inflow events respectively.

3.2. Nutrient output and load

3.2.1. Total nitrogen

Biochar (T), vegetation filter strips (VFS), biochar + vegetation filter strips (VFST) can significantly reduce the TN concentration in surface runoff (SR), but the effect of VSFT combination is the most obvious (Fig. 3a). In all rainfall events, the concentration of VFST90 was the lowest. Compared with CK, VFST90 decreased by 51.7 %, 48.6 % and 31.4 % respectively in three rainfall events. Compared with the changes of TN concentration in subsurface runoff (SSR), in all rainfall events, the application of biochar alone increased the TN concentration of SSR, VFS decreased the TN concentration of SSR, and the combined effect of VFST changed with the change of biochar application rate, while VFST30 and VFST60 reduced the TN concentration (Fig. 3b). All experiments had little effect with rainfall events and concentration gradients, and VFS was generally at the lowest TN concentration in all rainfall times.

Compared with CK, biochar (T), vegetation filter strips (VFS), biochar + vegetation filter strips (VFST) can significantly reduce the TN output load in surface runoff (SR) (Fig. 4a). And the effect of VFST30 was the most prominent, reducing the TN output load of 279.5 mg, 166.6 mg and 92.3 mg in the three rainfall events. Although VFS and T treatments alone also have an effect on reducing TN output load, their effect and stability are not as good as Vastin addition, the TN output load of VFS in the first two rainfall events was lower than that of T30 and T60, and higher than that of them in the third rainfall event.

For the reduction rate of TN output load, compared with CK, biochar (T), vegetation filter strips (VFS), biochar + vegetation filter strips (VFST) can significantly reduce the TN output load, but the reduction effect decreased with the increase of the event (Table 3). In all treatments, the TN load reduction rate of VFST30 was the highest, which was 59.9 %, 56.0 % and 45.0 % lower than that of CK in three rainfall events, respectively. The average reduction rate of VFST60 was also more than 50.3 %. In all experiments, T90 had the worst reduction efficiency of TN output load, with an average of only 14.5 %.

3.2.2. Total phosphorus

Among all rainfall events, VSFT had the best effect on reducing TP concentration in SR, and VSFT90 had the most obvious effect on reducing TP concentration. The reduction degrees of three rainfall events were 66.3 %, 67.3 % and 77.6 %, respectively (Fig. 5a). And it can maintain a low concentration fluctuation in the middle and late stages of each rainfall event. The effect of applying biochar alone in multiple rainfalls was slightly worse than that of VFST combination. At the beginning of all rainfall events, the difference between VFS and CK was not significant, but the TP concentration gradually decreased with the increase of rainfall. For the concentration curve

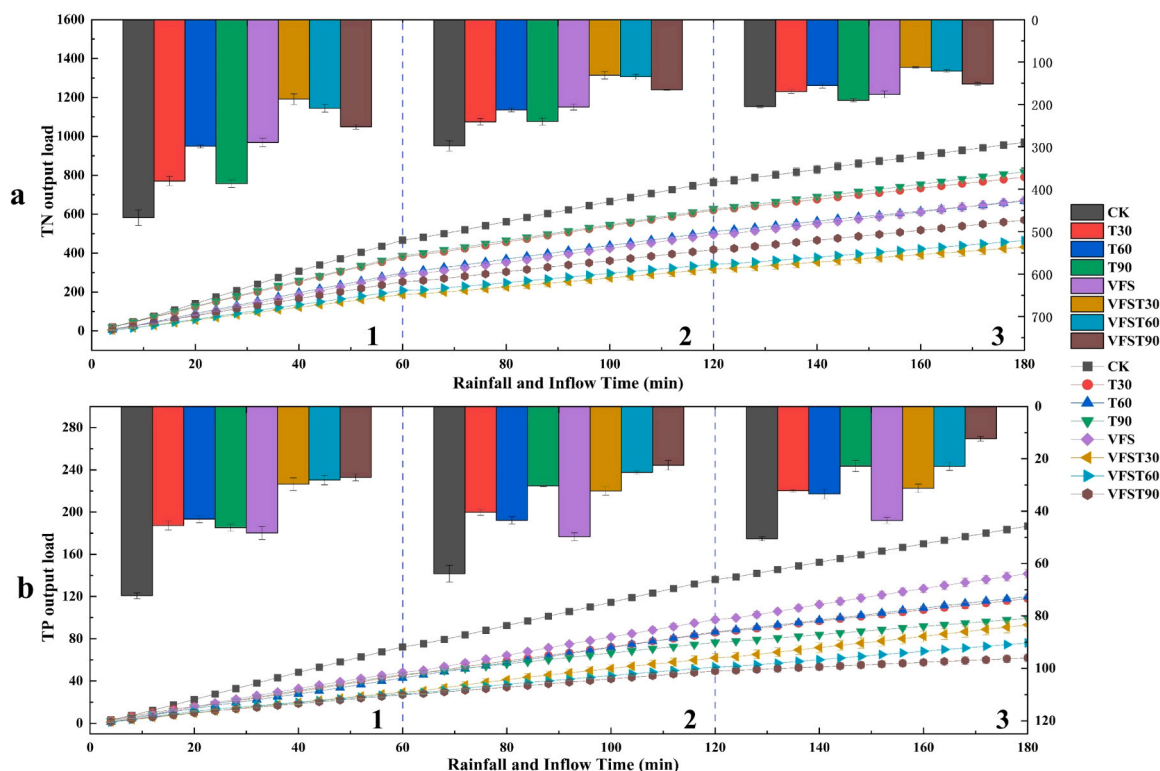


Fig. 4. The TN (a) and TP (b) output load of surface runoff (SR) under three consecutive rainfall and inflow events for Vegetative Filter Strips (VFS), individual application of biochar(T), and the coupled impact of Vegetative Filter Strips and biochar (VFST). The point line diagram represents the cumulative load of the three fields, and the histogram represents the rainfall load of a single field. The average of three replicates is shown with standard deviation bars. 1, 2 and 3 represent the first, second and third rainfall and inflow events respectively.

Table 3

TN output loads, reduction efficiency (RE), and average reduction efficiency (ARE) for each treatment for three rainfall events.

		CK	T30	T60	T90	VFS	VFST30	VFST60	VFST90
Event 1	TN output load	466.91	380.51	298.95	386.84	289.51	187.44	208.69	252.87
	RE		18.5 %	36.0 %	17.1 %	38.0 %	59.9 %	55.3 %	45.8 %
Event 2	TN output load	297.84	240.79	212.76	240.48	205.92	131.17	134.59	165.47
	RE		19.2 %	28.6 %	19.3 %	30.9 %	56.0 %	54.8 %	44.4 %
Event 3	TN output load	204.94	169.54	155.53	190.16	176.19	112.62	121.28	151.57
	RE		17.3 %	24.1 %	7.2 %	14.0 %	45.0 %	40.8 %	26.0 %
	ARE		18.3 %	29.6 %	14.5 %	27.6 %	53.6 %	50.3 %	38.7 %

of SSR, all experiments reduced the TP concentration of SSR (Fig. 5b). Compared with CK, the TP concentration of VFS or biochar alone fluctuated greatly during rainfall. VFST, especially VFST30, showed the best effect. It not only had the most significant effect on the reduction of TP concentration, which decreased by 58.8 %, 60.1 % and 57.5 % respectively in three rainfall events, but also showed higher stability with the increase of rainfall events.

For the TP output load of SR, the effect of VFST combination was the most significant (Fig. 4b). Among them, VFST90 had the best effect on reducing TP output load, which decreased by 45.2 mg, 41.4 mg and 38.3 mg in three rainfalls, respectively. Although VFS and biochar treatment alone also showed a decrease in TP output load, the effect gradually decreased in continuous rainfall. In addition, T90 is slightly higher than VFS in the first rainfall event and significantly lower than VFS in subsequent rainfall events.

For the reduction rate of TP output load, compared with CK, VFST90 had the highest TP reduction rate and the best reduction effect in the last rainfall, which decreased by 62.5 %, 64.8 % and 75.7 % respectively in the three rainfall events (Table 4). The reduction effect of VFST60 is also better, and the average reduction rate of TP output load is 58.9 %. In all experiments, VFS had the lower reduction efficiency of TP output load, with an average of only 23.0 %.

3.3. The relationship between runoff and nutrient characteristics, rainfall events, biochar

The correlation between rainfall events and biochar application and runoff and sediment yield characteristics of CK and VFS was analyzed (Fig. 6). Biochar showed a significant positive correlation with SR and a significant negative correlation with SSR. At the same

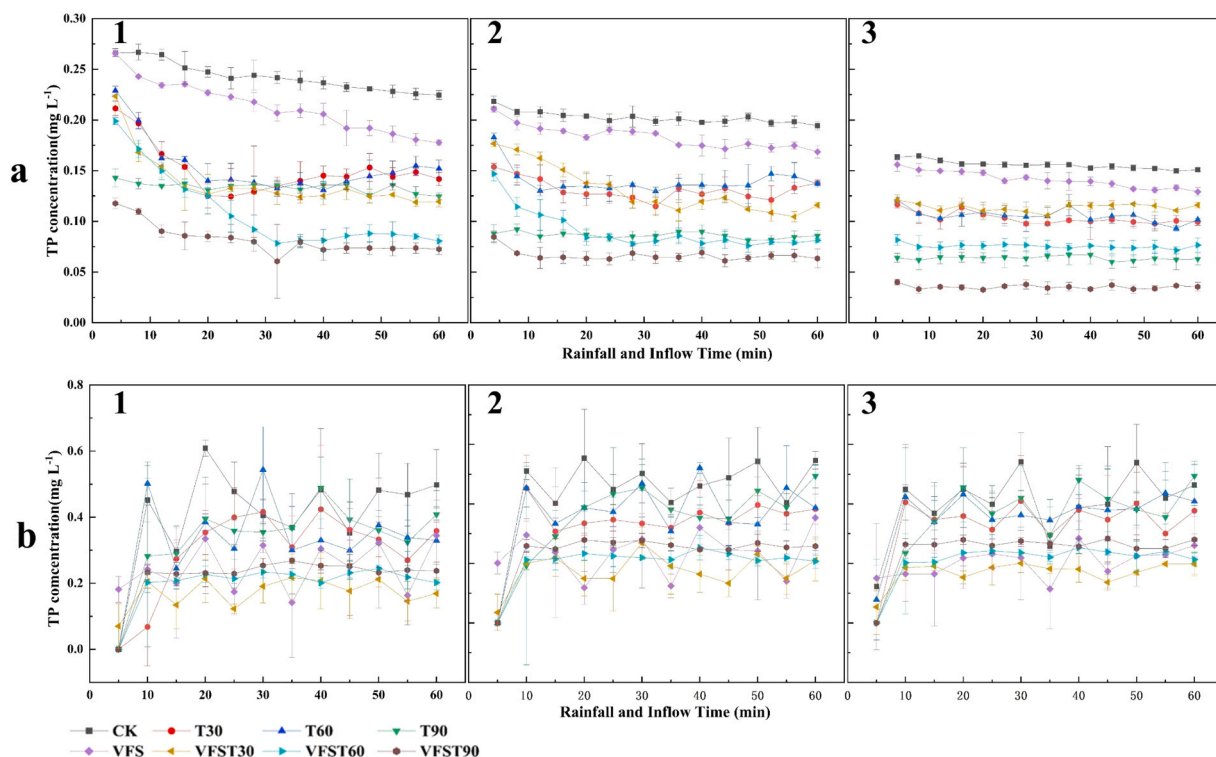


Fig. 5. The TP concentration of surface runoff (SR) (a) and subsurface runoff (SSR) (b) under three consecutive rainfall and inflow events for Vegetative Filter Strips (VFS), individual application of biochar(T), and the coupled impact of Vegetative Filter Strips and biochar (VFST). The average of three replicates is shown with standard deviation bars. 1, 2 and 3 represent the first, second and third rainfall and inflow events respectively.

Table 4

TP output loads, reduction efficiency (RE), and average reduction efficiency (ARE) for each treatment for three rainfall events.

		CK	T30	T60	T90	VFS	VFST30	VFST60	VFST90
Event 1	TP output load	72.30	45.44	43.01	46.26	48.33	29.67	28.09	27.11
	RE		37.2 %	40.5 %	36.0 %	33.2 %	59.0 %	61.1 %	62.5 %
Event 2	TP output load	63.86	40.40	43.47	30.30	49.75	32.24	25.16	22.48
	RE		36.7 %	31.9 %	52.6 %	22.1 %	49.5 %	60.6 %	64.8 %
Event 3	TP output load	50.59	32.15	33.39	22.73	43.56	31.24	22.80	12.30
	RE		36.4 %	34.0 %	55.1 %	13.9 %	38.2 %	54.9 %	75.7 %
	ARE		36.8 %	35.5 %	47.9 %	23.0 %	48.9 %	58.9 %	67.7 %

time, biochar significantly reduced the nutrient concentration and load in SR, and the reduction degree tended to increase with the increase of rainfall events. For CK and VFS, biochar application reduced the TP concentration of SSR and significantly reduced E_i (TP). The application of biochar in VFS increased the TP concentration of SSR and significantly reduced E_i (TP).

PLS-SEM was used to analyze the correlation between different treatment measures and rainfall time and nutrient concentration. The simulation results show that the model fits well with SRMR 0.9, as shown in Fig. 7. The results showed that Event had a significant negative impact on TN (SR) (path index: -0.860). BT had a significant negative effect on $Q_{(SSR)}$ and $TP_{(SR)}$ (path index: -0.847 , -0.837), and a significant positive effect on $TN_{(SSR)}$ (path index: 0.787). PL had a significant negative effect on $TN_{(SSR)}$ and $TP_{(SSR)}$ (path index: -0.736 , -0.928). The path of runoff is less significant to nutrient concentration, which indicates that vegetation cover, rainfall events and biochar directly affect the nutrient concentration of runoff.

4. Discussion

4.1. Effects of VFS and biochar on runoff process

The results demonstrated that vegetation filter strips (VFS) significantly reduce surface runoff (SR) rates while enhancing sub-surface runoff (SSR) rates. This effect is largely attributed to the ability of plant structures to intercept and redistribute rainfall. Canopy interception and stemflow capture reduce both the energy of falling raindrops and the overall depth of surface runoff (Wei et al., 2014).

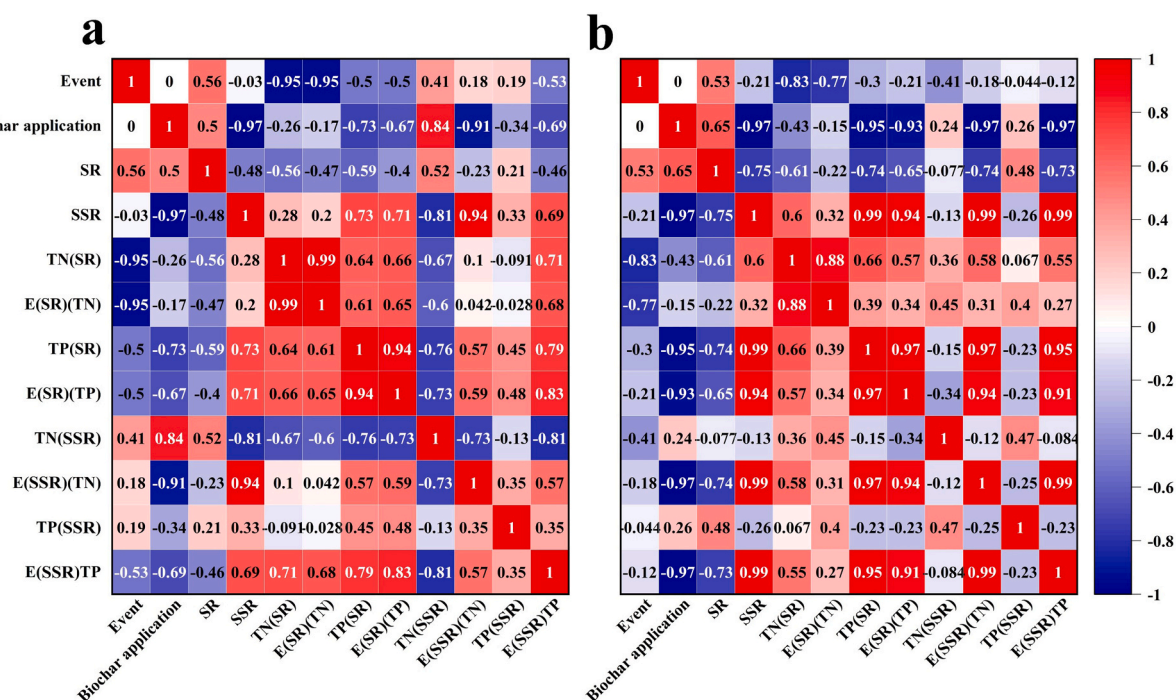


Fig. 6. Correlation analysis of (a) CK and (b) Vegetative Filter Strips (VFS) for runoff and nutrient characteristics, rainfall events, biochar. Correlation analysis of (a) CK and (b) Vegetative Filter Strips (VFS) for runoff and nutrient characteristics, rainfall events, biochar. Event represents rainfall event, Biochar application represents biochar application rate, SR represents surface runoff, SSR represents underground runoff, TN (SR) represents TN in surface runoff, E (SR) (TN) represents TN load in surface runoff, TP (SR) represents TP in surface runoff, E (SR) (TP) represents TP load in surface runoff, TN (SSR) represents TN in subsurface runoff, E (SSR) (TN) represents TN load in subsurface runoff, TP (SSR) represents TN in subsurface runoff, E (SSR) (TP) represents TP load in subsurface runoff.

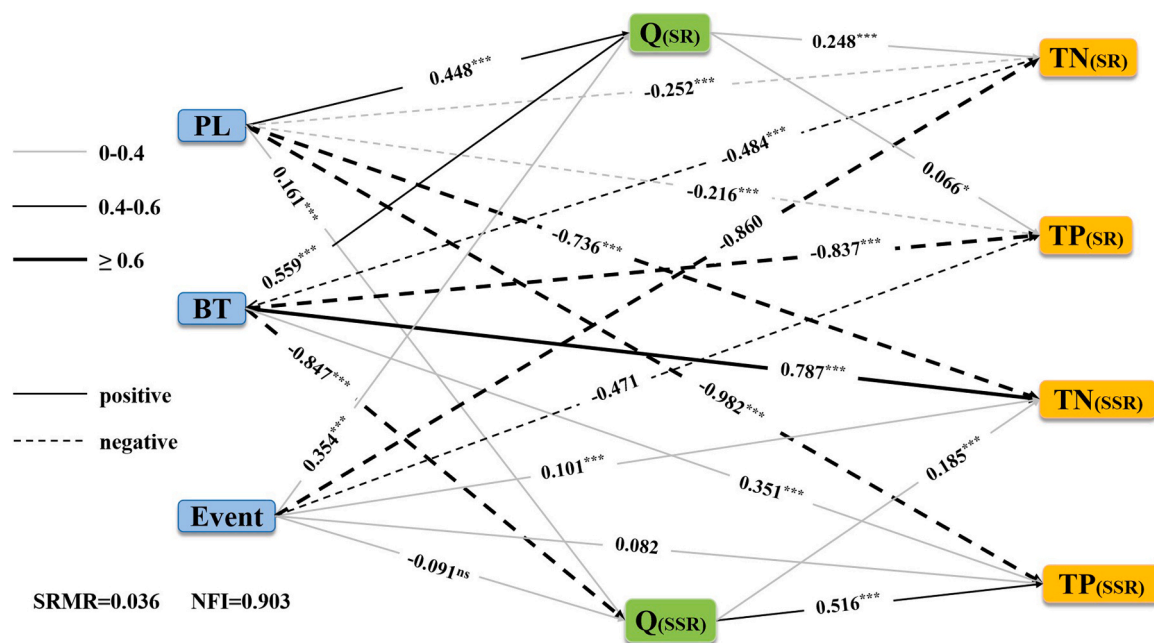


Fig. 7. The results of the partial least squares structural equation modeling. PL represents vegetation cover, BT represents biochar application, and Event represents different rainfall events. The numbers and asterisks next to the arrows represent the estimates and the correlation significance. The thickness of the arrow represents the significance of the path. Significant correlation ($p < 0.05$), high significant correlation ($p < 0.01$) and extremely significant correlation ($p < 0.001$) were represented by single-star (*), double-star (**) and three-star (***), respectively. Standardized root mean square residual (SRMR) and normalized fit index (NFI) represent model simulation indicators. (+) indicates a positive impact, (-) indicates a negative impact.

The influence of biochar on runoff rates, however, varies with the amount applied. When biochar is applied at 30 t ha^{-1} or 60 t ha^{-1} , it slightly reduces SR rates. In contrast, at 90 t ha^{-1} , biochar significantly increases SR rates. This finding contrasts with the results of Zhang et al. (2019) and Fan et al. (2023), possibly due to the differing interactions between biochar particles and the soil matrix at varying application rates (Masiello and Berhe, 2020). When VFS is combined with different biochar application rates, the effect on SR varies with increasing biochar dosage. This may result from the complex interplay between biochar, soil properties, and plant growth. Biochar promotes plant growth, which in turn enhances rainfall interception, thereby reducing rainwater splashing. Additionally, biochar can improve soil cohesion by stimulating root development (Gholamhadi et al., 2023), and it may also adsorb organic compounds released during root growth, further enhancing soil structure and fertility (Joseph et al., 2010).

The experimental results revealed that the application of biochar alone significantly reduced SSR rates. This effect may be attributed to biochar's ability to alter soil pore structure and significantly enhance saturated hydraulic conductivity (Ajayi et al., 2016). Similarly, applying biochar in combination with VFS also reduced SSR, though the coupling effect was less pronounced compared to biochar alone in reducing surface runoff. This can be explained by the inherent capacity of VFS to improve soil permeability. Vegetation roots enhance soil aggregate structure and porosity through interlacing, networking, and consolidation processes (Sadeghi et al., 2020). However, biochar tends to become locally saturated near the roots, which contributes to increased SSR. The presence of macropores around the roots promotes preferential flow pathways, accelerating the generation of SSR (Weiler and Naef, 2003). This suggests that while both biochar and VFS enhance soil permeability, their combined application may not provide as substantial a reduction in surface runoff as biochar alone due to these root-associated dynamics.

4.2. Effects of VFS and biochar on runoff nutrient output

The results showed that the application of biochar alone significantly reduced the nutrient density, which was similar to the conclusion of Jin et al. (2022). First, vegetation itself can absorb and degrade nitrogen and phosphorus in fertilizers through rhizomes (Gabriel and Quemada, 2011; Wu et al., 2021). Secondly, VFS reduces the flow friction slope of water flow, thus reducing the flow force and promoting the absorption of nutrients in runoff by plants (Zheng et al., 2020). Finally, VFS not only directly absorbs the nutrients exported from runoff, but also promotes the development and activity of rhizosphere microorganisms and aerobic microorganisms. These microorganisms can absorb and accumulate nutrients in runoff and induce plant degradation. (Moore et al., 2017; Liu et al., 2019).

The application of biochar alone reduced the TN concentration in SR, because the ion exchange capacity and the presence of active functional groups on the surface of biochar were beneficial to the adsorption of nutrients (Lee and Park, 2020). Biochar can increase soil pH, cation exchange capacity (CEC) and water holding capacity, thereby further reducing the nitrogen loss caused by SR (Sim

et al., 2021). However, the TN concentration in SSR increased as a whole. This is because the biochar itself contains a large amount of nitrogen. For example, (Zhang et al., 2019) study measured the total nitrogen content of biochar to be 1543.83 ± 1346.68 ppm, which leads to the loss of nitrogen in the SSR with more complex water flow pathways. The amount of nitrogen lost exceeds the absorption capacity of excess biochar for nitrogen. Our study also showed that the application of biochar alone reduced TP concentrations in SR and SSR. Because biochar enhanced the negative ion exchange capacity that affects the interaction between phosphorus and cations, thereby increasing the availability of phosphorus and reducing the loss of soluble phosphorus (Zhao et al., 2023). This finding contrasts with the research of Fan et al. (2023), which reported that biochar application had no significant impact on the TP in SSR within yellow soil regions. The discrepancy may stem from differences in soil properties between the study areas. Unlike yellow soil, red soil typically exhibits a higher bulk density, and the interactions between biochar particles and the soil matrix vary significantly. These unique characteristics of red soil may facilitate the adsorption of TP in SSR more effectively than in other soil types.

Compared with VFS, the coupling effect of VFS and different amounts of biochar significantly reduced the nutrient concentration in SR and SSR, which was because the application of biochar increased the soil pH and the retention rate of soil moisture and nutrients (Sun et al., 2017). The study of (Murtaza et al., 2023) showed that the application of biochar improved the matrix characteristics of soil (effective nitrogen increased by 17 %, total carbon increased by 13 %), and promoted the morphology of plant roots, buds and leaves. (Obia et al., 2018) showed that compared with untreated plants, the application of biochar increased the root length, root surface area and biomass of roots, stems and leaves by twice. At the same time, biochar significantly increased the net photosynthetic rate, transpiration rate, stomatal conductance and water use efficiency of plants at maturity. These factors make biochar enhance the absorption capacity of plants to nutrients (Gao et al., 2017; Masiello and Berhe, 2020). Combined with the characterization analysis of soil after biochar application (Fig. 8), a series of complex reactions may also occur between plant roots and biochar. During rainfall, biochar destroyed soil aggregates and formed several large pores on the soil surface. Plant root hairs may contact pores and adsorb biochar particles. Root hairs can enter large pores filled with water and can also bind to the surface of biochar (Joseph et al., 2010). Biochar is adsorbed near the roots of vegetation. The macropore structure around the roots can change the flow path and speed of surface water, leading to the redistribution of water in the soil profile, thereby increasing the residence time of water in the soil, which is conducive to the adsorption of nutrients by the roots.

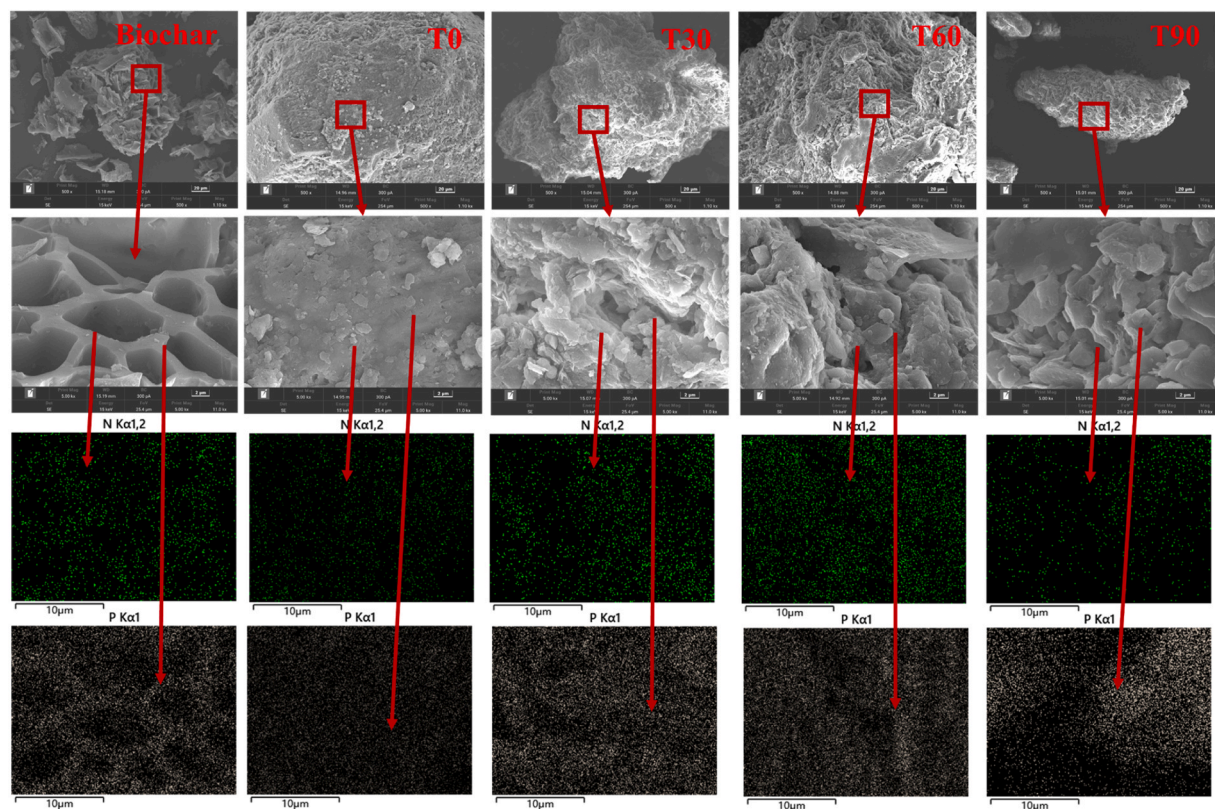


Fig. 8. Scanning electron microscopy images and energy dispersive X-ray spectroscopy surface scans were performed on biochar, soil aggregates, and soil aggregates treated with different amounts of biochar. In the surface scanning, the green dots represent the nitrogen (N) element, the white dots represent the semi-quantitative analysis of phosphorus (P), and the higher number of red dots indicates the presence of more carbon elements.

4.3. Effects of VFS and biochar on runoff nutrient loading characteristics

In this study, VFS reduced the output load of TN and TP. In addition to the adsorption of nutrients in SR by plants, VFS also delayed the production of SR to a considerable extent, which was similar to the results of Zhao et al. (2014) and Blanus and Hadley. (2019). This is because the morphology and biomass of herbaceous plants are important factors affecting the reduction efficiency. Physical adsorption is the most direct way to reduce pollutants (Hu et al., 2023). The total amount of rainwater captured by plants through canopy interception and stem flow may help to reduce raindrop energy and total runoff depth, thereby reducing nutrient loss (Wei et al., 2014). The roots of plants also enhance the soil matrix by physically binding and connecting soil particles on the soil surface, thereby enhancing the absorption capacity of nutrients in the runoff.

The application of biochar alone generally reduced the output loads of TN and TP in SR, which is like the conclusion (Yin et al., 2023). This is because biochar-modified soil can adsorb NO_3^- through its anion exchange sites, thereby reducing N loss and increasing NO_3^- retention (Yao et al., 2012). At the same time, biochar has the ability to retain inorganic nitrogen ions, so it reduces nitrogen loss in runoff (Steiner et al., 2008). In our study, T90 significantly increased the SR rate, but its TP output load was the lowest. This is because the possible mechanism of biochar affecting phosphorus availability is the change of soil pH value and its effect on the interaction between phosphorus and other cations, as well as the enhanced retention through anion exchange and phosphorus precipitation (Atkinson et al., 2010). The biochar at this dosage has the strongest adsorption capacity for phosphorus. SEM analysis (Fig. 8) showed that 90 t / ha biochar caused significant damage to soil aggregates during rainfall. The fragmentation, wedging or dissolution of biochar in soil aggregates leads to the formation of multi-layered, hollow and weakly structured surfaces, which may lead to the rapid horizontal flow of rainwater that cannot be retained. However, its porous structure and larger surface area make biochar-soil aggregates adsorb more nutrients from runoff. At the same time, in the natural environment, phosphorus is strongly adsorbed on the surface of iron (III) - (hydrogen) oxides in soil (Jaisi et al., 2010). The addition of biochar reduced the amount of phosphorus adsorbed on hydrated iron (the most effective iron oxide for phosphorus adsorption).

Compared with VFS alone, VFST significantly reduced the nitrogen and phosphorus load of SR. This is because biochar is rich in oxygen-containing functional groups (Keesstra et al., 2016). This feature makes biochar have good ion exchange adsorption capacity and can effectively adsorb soluble N and P in soil. At the same time, a high proportion of biochar can adsorb nitrate nitrogen, weaken nitrification in soil, and further reduce nitrogen content, resulting in reduced nitrogen loss with runoff Yuan and Che.(2022). In addition, the combined application of fertilizer and biochar can improve soil structure, promote plant root growth, and improve plant nitrogen utilization, thereby reducing the loss of nitrogen and phosphorus (Liu et al., 2014; Palansooriya et al., 2019). Therefore, in the case of co-application of fertilizer and biochar, the amount of fertilizer can be appropriately reduced. Therefore, in the case of co-application of fertilizer and biochar, the amount of fertilizer can be appropriately reduced. Biochar input can change soil physical and chemical properties, including pH value, capacity weight, CEC, water retention and biological activity, thus affecting soil nutrient retention. Biochar can significantly increase soil resin soluble phosphorus and inorganic soluble phosphorus, thus promoting the increase of nutrient available phosphorus in soil. At the same time, biochar may convert insoluble and unavailable nitrogen and phosphorus into plant-available forms during rainfall (Lee et al., 2018). In addition, the formation of biochar will produce many functional groups and positively charged ions, which can effectively fix and remove nitrogen ions through the ion exchange process in the environment (Zhao et al., 2023). Biochar also enhances the negative ion exchange capacity that affects the interaction between phosphorus and cations (Zhang et al., 2021). Combined with the characterization analysis of biochar (Fig. 8), a series of complex reactions may also occur between plant roots and biochar. In the process of rainfall, biochar destroyed soil aggregates and formed several large pores on the soil surface. Plant root hairs may contact pores and adsorb biochar particles. Root hairs can enter large pores filled with water or bind to the surface of biochar. In addition, the porous structure of biochar itself contributes to the retention of water and the enhancement of water-vapor cycle, which is a key factor determining the adsorption capacity of biochar (Zhang et al., 2022). These factors together reduce the nutrient load of surface runoff.

4.4. Nutrient output characteristics of continuous rainfall and inflow

Our study shows that different rainfall events in the process of continuous rainfall have different results on nutrient reduction rate. This is because with the increase of rainfall events, the infiltration rate of soil will increase, and its hydraulic conductivity may also change (Wang et al., 2018). Changes in the surface charge properties of the soil also lead to changes in the interaction between soil particles. At the same time, frequent rainfall events affect the overall surface roughness of the soil, resulting in different degrees of soil absorption of runoff nutrients (Shen et al., 2023).

For VFS, we observed that the effectiveness of nutrient reduction gradually weakened as the event progressed, but the overall nitrogen reduction efficiency was better. This is attributed to the fact that the drying-rewetting alternation caused by continuous rainstorms satisfies the moderate water conditions required for ammonia volatilization loss. Soil water evaporation can promote the movement of soluble nitrogen on the soil surface, thereby increasing the storm runoff loss of soluble nitrogen (Zhang et al., 2020). On the other hand, due to the gap between the canopy leaves of the vegetation, the slope is divided into several channels with rainfall and upper catchment events. Each channel forms a natural flow channel above the slope, which increases the flow connectivity of the VFS and increases the runoff. This explains to some extent the reason why the efficiency of nutrient reduction gradually decreases with the number of events. Compared with VFS, the nutrient reduction efficiency under the coupling of VFS and biochar had less effect with the increase of rainfall events, and the nutrient load was reduced to the greatest extent. At the same time, high proportion of biochar addition can adsorb nitrate in soil, weaken nitrification in soil, further reduce soil TN concentration, and reduce TN loss with runoff (Liu et al., 2019). In addition, the combination of fertilizer and biochar can improve soil structure, promote plant root growth, and

improve plant nitrogen utilization. It can be speculated that in addition to the reason of biochar itself, biochar may also assist the stem flow of plants to reserve precipitation, greatly wet the soil around the roots, and adsorb soil particles as a whole, thereby improving the soil structure (Wei et al., 2014), resulting in a more consistent trend in the process of runoff production, thus overcoming the influence of the increase of nutrient output with the increase of rainfall events after the separate layout of VFS, and reducing the nutrients in runoff to the greatest extent.

4.5. Optimum application amount of biochar

Our correlation analysis showed that the synergistic effect of VFS and biochar changed the correlation between runoff characteristics and nutrient output, and reduced the non-point source pollution of continuous rainfall events. This is similar to the results of (Zhao et al., 2023), that is, the application of biochar reduced the concentration of TN and TP in the runoff. Our study also showed that the application of biochar in VFS also significantly reduced runoff and further reduced nutrient load. The SEM-PLS equation showed that runoff positively affected the nutrient concentration of runoff, while the application amount of biochar and the number of rainfall events significantly affected the nutrient concentration of runoff, indicating that the application of biochar could significantly reduce the nutrient loss of continuous rainfall events. This is because the runoff nitrogen of plant slope mainly appears in the form of dissolved part (Wu et al., 2022). Biochar aggravates the interaction between runoff and surface soil, making the soil adsorb more nutrients in this process (Wang et al., 2021). Considering the actual benefits and costs associated with biochar, we recommend the rational application of biochar as an improver in soil vegetation management practices to improve soil quality, reduce nitrogen and phosphorus loss, and promote plant growth. This strategic application of biochar can not only improve environmental quality, but also support sustainable agricultural practices by stabilizing nutrient cycling in ecosystems.

5. Conclusion

In this study, the nutrient concentration and load characteristics under the single effect and coupling effect of VFS or biochar were studied by simulating continuous rainfall and inflow flume experiments. VFS and biochar affected all the research variables of nutrients, and the application of 30–90 t / ha biochar in VFS could significantly reduce the non-point source pollution in the riparian zone. At the same time, the application of biochar weakened and stabilized the nutrient loss load of continuous rainfall events. VFS-biochar coupling minimizes the net loss of nutrients through surface interception and seepage adsorption. Although SSR reduction was weak, the core ability of biochar to adsorb dissolved nutrients in the seepage path was not weakened. Consequently, the combined application of VFS and biochar appears to be an effective strategy for enhancing nutrient flow management in riparian zones and into basins. This technology has the synergistic benefits of environment, economy and ecology, and can be applied to the fields of agricultural green transformation and urban water environment governance. In the future, further research is needed to increase the mechanism of adsorption of polymorphic pollutants. In particular, the long-term effects under natural rainfall conditions and the adaptability of different soil-crop systems.

Author Statement

We declare that this manuscript is original, has not been published before and is not currently being considered for publication elsewhere.

We confirm that the manuscript has been read and approved by all named authors and that there are no other persons who satisfied the criteria for authorship but are not listed. We further confirm that the order of authors listed in the manuscript has been approved by all of us.

We understand that the Corresponding Author is the sole contact for the Editorial process. He is responsible for communicating with the other authors about progress, submissions of revisions and final approval of proofs.

CRedit authorship contribution statement

Qinxue Xu: Writing – review & editing, Writing – original draft, Methodology, Funding acquisition, Conceptualization. **Xiuxian Zhu:** Writing – review & editing, Writing – original draft, Methodology, Formal analysis, Conceptualization. **Xinrui Yu:** Methodology, Conceptualization. **Liman Ao:** Methodology, Conceptualization. **Jiawei Li:** Writing – review & editing, Writing – original draft, Formal analysis, Data curation, Conceptualization. **Shuaipu Zhang:** Methodology, Conceptualization.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

This study was supported by the Guangxi Key R&D Program [Guike-AB22035075], the Guangxi Science and Technology Program [Guike- AD25069074].

Data availability

The data that has been used is confidential.

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